Non-dominated Sorting

1 Non-dominated Sorting

Let $\mathbb{P} = \{P_1, P_2, \ldots, P_N\}$ be a population consisting of N points in an M-dimensional space. A point P_i in an M-dimensional space is represented as $P_i = \langle P_{i_1}, P_{i_2}, \ldots, P_{i_M} \rangle$. **Without loss of generality, we assume that all objectives need to be minimized**. Under these conditions, Pareto dominance, or simply dominance, is defined as follows:

Definition 1 (**Dominance**). *A point $P_i = \langle P_{i_1}, P_{i_2}, \ldots, P_{i_M} \rangle$ dominates a point $P_j = \langle P_{j_1}, P_{j_2}, \ldots, P_{j_M} \rangle$, written as $P_i \prec P_j$, if the following two conditions are satisfied:*

- *Point P_i is **better than or equal to** point P_j for all the objectives. Mathematically, $\forall m \in \{1, 2, \ldots, M\}$ $P_{i_m} \leq P_{j_m}$*
- *Point P_i is **strictly better than** point P_j for at least one objective. Mathematically, $\exists m \in \{1, 2, \ldots, M\}$ $P_{i_m} < P_{j_m}$*

When the point P_i does not dominate the point P_j, it is denoted as $P_i \not\prec P_j$. When point P_i does not dominate point P_j and point P_j does not dominate point P_i, we say that P_i and P_j are *non-dominated* with each other, *i.e.*, P_i and P_j are non-dominated if $(P_i \not\prec P_j) \wedge (P_j \not\prec P_i)$.

Definition 2 (**Front**). *A set of points that are non-dominated with each other constitutes a front.*

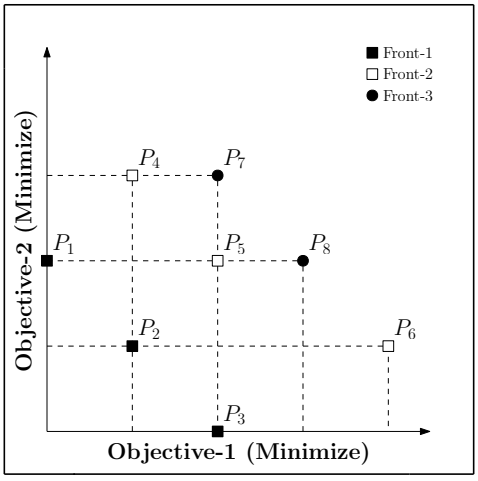


Figure 1: Population having eight points in two dimensional space.

Definition 3 (**Non-dominated Sorting**). *Non-dominated sorting is defined as the procedure that produces a set of* fronts $\mathbb{F} = \{F_1, F_2, \ldots, F_K\}$ *where $1 \leq K \leq N$, such that the following conditions hold:*

- *Union of the points in all the fronts is same as the population*
  - $\bigcup_{k \geq 1} F_k = \mathbb{P}$
- *The fronts are disjoint*
  - $\forall i \neq j,\ F_i \cap F_j = \emptyset$
- *All the points in a particular front are non-dominated with each other*

- $\forall k \geq 1,\ \forall P, Q \in F_k : (P \not\prec Q) \wedge (Q \not\prec P)$

- *The first front consists of points that are not dominated by any point*
  - $\forall P \in F_1,\ \nexists Q \in \mathbb{P} : Q \prec P$;

- *All the points in a particular front are dominated by at least one of the points in its preceding front*
  - $\forall P \in F_k, k > 1\ \exists Q \in F_{k-1} : Q \prec P$.

Assume the cardinality of a front F_k be n_k, so $N = \sum_{k=1}^{K} n_k$. Figure 1 shows the set of eight points in two-dimensional space. These eight points are divided into three fronts such that $n_1 = 3$, $n_2 = 3$, and $n_3 = 2$.

2 Properties of Dominance Relation

Here, we explore the properties of the dominance relation as discussed in [1]. Specifically, we delve into the binary relation properties [3] of the dominance operator.

1. **Irreflexive:** As per the second condition of Definition 1, a point $P = \langle P_1, P_2, \ldots, P_M \rangle$ cannot dominate itself. Thus, the dominance relation is **not reflexive**.

2. **Asymmetric:** As per the definition of the dominance (Definition 1), if $P \prec Q$, then point Q cannot dominate P. Thus, the dominance relation is **asymmetric**.

3. **Antisymmetric:** As the dominance relation is not symmetric, it cannot be **antisymmetric**.

4. **Transitivity:** Let there be three points $P = \langle P_1, P_2, \ldots, P_M \rangle$, $Q = \langle Q_1, Q_2, \ldots, Q_M \rangle$ and $R = \langle R_1, R_2, \ldots, R_M \rangle$ in the population such that $P \prec Q$ and $Q \prec R$.

   Point P dominates Q if both the following conditions are satisfied:

   - $\forall m \in \{1, 2, \ldots, M\}\ P_m \leq Q_m$
   - $\exists m \in \{1, 2, \ldots, M\}\ P_m < Q_m$.

   Similarly, point Q dominates R if both the following conditions are satisfied:

   - $\forall m \in \{1, 2, \ldots, M\}\ Q_m \leq R_m$
   - $\exists m \in \{1, 2, \ldots, M\}\ Q_m < R_m$.

   Observation 1: Given points Q and R, if $Q \prec R$, it implies there exists at least one objective m for which $Q_m < R_m$. For this particular objective m, $P_m \leq Q_m$. Combining these two statements, $P_m \leq Q_m$ and $Q_m < R_m$, we get $P_m < R_m$. Thus, for at least one objective m, $P_m < R_m$.

   Observation 2: Given points Q and R, if $Q \prec R$, then for all the objectives m, $Q_m \leq R_m$. Similarly, point $P \prec Q$, it means for all the objectives m, $P_m \leq Q_m$. Combining these two statements, $P_m \leq Q_m$ and $Q_m \leq R_m$, we get $P_m \leq R_m$. Thus, for all the objectives m, $P_m \leq R_m$.

   Combining observations 1 and 2, we get the following

   - $\forall m \in \{1, 2, \ldots, M\}\ P_m \leq R_m$ [**Observation 2**]
   - $\exists m \in \{1, 2, \ldots, M\}\ P_m < R_m$ [**Observation 1**]

   Thus, $P \prec R$. Hence, the dominance relation is **transitive**.

5. **Another Property:** If point P does not dominate Q, this does not imply that Q dominates P.

   Proof. Point $P \prec Q$, if both the following conditions (S1 and S2) are satisfied:

   - **S1:** $\forall m : P_m \leq Q_m$
   - **S2:** $\exists m : P_m < Q_m$

i.e., $P \prec Q \Rightarrow \text{S1} \wedge \text{S2}$. Therefore,

$$\begin{aligned} P \not\prec Q &\Rightarrow \neg\left(\text{S1} \wedge \text{S2}\right) \\ &\Rightarrow \neg\text{S1} \vee \neg\text{S1} \\ &\Rightarrow \neg\left(\forall m : P_m \leq Q_m\right) \vee \neg\left(\exists m : P_m < Q_m\right) \\ &\Rightarrow \underbrace{\left(\exists m : P_m > Q_m\right)}_{\text{S3}} \vee \underbrace{\left(\forall m : P_m \geq Q_m\right)}_{\text{S4}} \end{aligned} \tag{1}$$

From Equation (1), we can say that P does not dominate Q, if any one or both the conditions (S3 and S4) are satisfied. Thus, if $P \not\prec Q$, one of the following conditions holds.

- **Case 1:** $\text{S3} \wedge \text{S4}$
- **Case 2:** $\text{S3} \wedge \neg\text{S4}$
- **Case 3:** $\neg\text{S3} \wedge \text{S4}$

Case 1:

$$\begin{aligned} &\text{S3} \wedge \text{S4} \\ &\Rightarrow \left(\exists m : P_m > Q_m\right) \wedge \left(\forall m : P_m \geq Q_m\right) \\ &\Rightarrow Q \prec P \end{aligned} \tag{2}$$

Case 2:

$$\begin{aligned} &\text{S3} \wedge \neg\text{S4} \\ &\Rightarrow \left(\exists m : P_m > Q_m\right) \wedge \neg\left(\forall m : P_m \geq Q_m\right) \\ &\Rightarrow \left(\exists m : P_m > Q_m\right) \wedge \left(\exists m : P_m < Q_m\right) \\ &\Rightarrow P \text{ and } Q \text{ are non-dominated} \end{aligned} \tag{3}$$

Case 3:

$$\begin{aligned} &\neg\text{S3} \wedge \text{S4} \\ &\Rightarrow \neg\left(\exists m : P_m > Q_m\right) \wedge \left(\forall m : P_m \geq Q_m\right) \\ &\Rightarrow \left(\forall m : P_m \leq Q_m\right) \wedge \left(\forall m : P_m \geq Q_m\right) \\ &\Rightarrow P \text{ and } Q \text{ have the same objective vector} \end{aligned} \tag{4}$$

Thus, when $P \not\prec Q$, it does not always means that $Q \prec P$. This is one of the possibility. In this case, actually three possibilities are there.

- Q dominates P
- P and Q are non-dominated.
- P and Q have the same objective vector.

□

3 Domination Count

Definition 4 (**Domination Count**). *The domination count of a point $P \in \mathbb{P}$ is the number of points in the population that dominates P [2]. It is denoted as $\texttt{dom_count}_P$.*
Mathematically, $\texttt{dom_count}_P = |\{Q | Q \in \mathbb{P} \wedge Q \prec P\}|$ where $|\cdot|$ denotes the cardinality of a set.

Example 1. *Consider the population shown in Figure 1. Now we calculate the domination count of each point in the population. For this purpose, we need to find that point P is dominated by how many points in the population. The domination count of the entire population is shown in Table 1.*

Theorem 1. *The dominance count of a point $P \in \mathbb{P}$ cannot be more than $N - 1$.*

Proof. There are N points in the population $\mathbb{P}$. A point $P \in \mathbb{P}$ does not dominate itself. So a point can only be dominated by the remaining $N - 1$ points. Thus, the dominance count of a point $P \in \mathbb{P}$ cannot be more than $N - 1$. □

Table 1: Domination count of the points in the population shown in Figure 1.

Point	Dominated By	Domination Count
P_1	$\emptyset$	0
P_2	$\emptyset$	0
P_3	$\emptyset$	0
P_4	$\{P_1, P_2\}$	2
P_5	$\{P_1, P_2, P_3\}$	3
P_6	$\{P_2, P_3\}$	2
P_7	$\{P_1, P_2, P_3, P_4, P_5\}$	5
P_8	$\{P_1, P_2, P_3, P_5\}$	4

3.1 Maximum/Minimum Value of Domination Count of a Point $P \in F_k$

Here, we discuss the maximum and minimum value of the domination count for a point $P \in F_k$, $1 \leq k \leq K$. The domination count of a point $P \in F_k$ represents the number of points that can dominate P. Hence, we aim to determine the maximum and minimum number of points that can dominate a given point P.

3.1.1 Maximum Value of Domination Count

As per the definition of non-dominated sorting [fifth condition of the Definition 3], a point $P \in F_k$ is dominated by **at least** one point in its preceding front F_{k-1}. It means P can also be dominated by **all the** points of its preceding front F_{k-1}, *i.e.*, $\forall Q \in F_{k-1} : Q \prec P$.

All the points of front F_{k-1} are dominated by **at least** one point in its preceding front F_{k-2}. It means all the points of F_{k-1} can also be dominated by **all the** points of its preceding front F_{k-2}, *i.e.*, $\forall R \in F_{k-2}, \forall Q \in F_{k-1} : R \prec Q$. As the dominance relation is transitive, so point P can also be dominated by **all the** points of front F_{k-2}, *i.e.*,

$$\begin{aligned}(\forall R \in F_{k-2}, \forall Q \in F_{k-1} : R \prec Q) \wedge \\ (\forall Q \in F_{k-1} : Q \prec P) \Rightarrow (\forall R \in F_{k-2} : R \prec P)\end{aligned} \tag{5}$$

All the points of front F_{k-2} are dominated by **at least** one point in its preceding front F_{k-3}. It means all the points of F_{k-2} can also be dominated by **all the** points of its preceding front F_{k-3}. *i.e.*, $\forall S \in F_{k-3}, \forall R \in F_{k-2} : S \prec R$. As the dominance relation is transitive, so point P can also be dominated by **all the** points of front F_{k-3}, *i.e.*,

$$\begin{aligned}(\forall S \in F_{k-3}, \forall R \in F_{k-2} : S \prec R) \wedge \\ (\forall R \in F_{k-2}, \forall Q \in F_{k-1} : R \prec Q) \wedge \\ (\forall Q \in F_{k-1} : Q \prec P) \Rightarrow (\forall S \in F_{k-3} : S \prec P)\end{aligned} \tag{6}$$

In a similar manner, point P can also be dominated by **all the** points of fronts $F_{k-4}, F_{k-5}, \ldots, F_1$. Thus, the point P can be dominated by all the points of fronts $F_{k-1}, F_{k-2}, \ldots, F_1$. Thus, the maximum number of points that can dominate a point $P \in F_k$ is the sum of the cardinality of all its preceding fronts $F_{k-1}, F_{k-2}, \ldots, F_1$, and is obtained by Equation (7).

$$\begin{aligned}\texttt{Domination_count}_{\texttt{max}} &= n_{k-1} + n_{k-2} + \cdots + n_1 \\ &= \sum_{i=1}^{k-1} n_i\end{aligned} \tag{7}$$

The number of points that can dominate a point $P \in F_k$ is basically the domination count of P. Thus, the maximum value of the domination count of a point $P \in F_k$ is given by Equation (7).

3.1.2 Minimum Value of Domination Count

As per the definition of non-dominated sorting [fifth condition of the Definition 3], a point $P \in F_k$ is dominated by **at least** one point in its preceding front F_{k-1}. It means P can be dominated by **exactly one** point (say $Q \in F_{k-1}$) of its preceding front F_{k-1}.

Point $Q \in F_{k-1}$ is dominated by **at least** one point of it preceding front F_{k-2}. It means Q can be dominated by **exactly one** point (say $R \in F_{k-2}$) of its preceding front F_{k-2}. As the dominance relation is transitive, so point P can also be dominated by point R, *i.e.*, $(R \prec Q) \wedge (Q \prec P) \Rightarrow R \prec P$.

Point $R \in F_{k-2}$ is dominated by **at least** one point of it preceding front F_{k-3}. It means R can be dominated by **exactly one** point (say $S \in F_{k-3}$) of its preceding front F_{k-3}. As the dominance relation is transitive, so point P can also be dominated by point S, *i.e.*, $(S \prec R) \wedge (R \prec Q) \wedge (Q \prec P) \Rightarrow S \prec P$.

In a similar manner, point P can be dominated by **exactly one** point of fronts $F_{k-4}, F_{k-5}, \ldots, F_1$. Thus, the point P can be dominated by exactly one point of fronts $F_{k-1}, F_{k-2}, \ldots, F_1$. Thus, the minimum number of points that can dominate a point $P \in F_k$ is obtained by Equation (8).

$$\begin{aligned} \texttt{Domination_count}_{\texttt{min}} &= \underbrace{1+1+\cdots+1}_{\text{One point of fronts } F_{k-1},F_{k-2},\ldots,F_1} \\ &= k-1 \end{aligned} \tag{8}$$

The number of points that can dominate a point $P \in F_k$ is basically the domination count of P. Thus, the minimum value of the domination count of a point $P \in F_k$ is given by Equation (8).

4 Dominance Set

Definition 5 (**Dominance Set**). *The dominance set of a point $P\mathbb{P}$ is the set of points in the population $\mathbb{P}$ that are dominated by P. It is denoted as $\mathbb{D}_P$.*
Mathematically, $\mathbb{D}_P = \{Q | Q \in \mathbb{P}, P \prec Q\}$.

Example 2. *Consider the population shown in Figure 1. Now we calculate the dominance set of each point in the population. For this purpose, we need to find the set of points dominated by point P in the population. The dominance set of the entire population is shown in Table 2.*

Table 2: Dominance set of the points in the population shown in Figure 1.

Point	Dominance Set
P_1	$\{P_4, P_5, P_7, P_8\}$
P_2	$\{P_4, P_5, P_6, P_7, P_8\}$
P_3	$\{P_5, P_6, P_7, P_8\}$
P_4	$\{P_7\}$
P_5	$\{P_7, P_8\}$
P_6	$\emptyset$
P_7	$\emptyset$
P_8	$\emptyset$

Theorem 2. *The maximum cardinality of the dominance set of a point $P \in \mathbb{P}$ cannot be more than $N-1$.*

Proof. There are N points in the population $\mathbb{P}$. A point $P \in \mathbb{P}$ does not dominate itself. So a point can dominate the remaining $N-1$ points. Thus, the maximum cardinality of the dominance set of a point $P \in \mathbb{P}$ cannot be more than $N-1$. □

4.1 Maximum/Minimum Cardinality of Dominance Set

Here, discuss the maximum and minimum cardinality of the dominance set of a point $P \in F_k$, $1 \leq k \leq K$. The number of points dominated by a point $P \in F_k$ is basically the cardinality of dominance set of P. So we obtain the maximum and minimum number of points dominated by P.

4.1.1 Maximum Cardinality of Dominance Set

As per the definition of non-dominated sorting [fifth condition of the Definition 3], a point $P \in F_k$ is dominated by **at least** one point in its preceding front F_{k-1}. It means P can also be dominated by **all the** points of its preceding front F_{k-1}. Also, all the points of a particular front can be dominated by the same point in its preceding

front as there is no such restrictions. Hence, all the points of front F_{k+1} can be dominated by a point $P \in F_k$, *i.e.*, $\forall P' \in F_{k+1} : P \prec P'$.

Similarly, all the points of front F_{k+2} can be dominated by a point $P' \in F_{k+1}$, *i.e.*, $P' \in F_{k+1}, \forall P'' \in F_{k+2} : P' \prec P''$. As dominance relationship is transitive, so all the points of front F_{k+2} can be dominated by point P, *i.e.*,

$$\begin{aligned}(P' \in F_{k+1}, \forall P'' \in F_{k+2} : P' \prec P'') \wedge \\ (\forall P' \in F_{k+1} : p \prec P') \Rightarrow (\forall P'' \in F_{k+2} : P \prec P'')\end{aligned} \tag{9}$$

In a similar manner, all the points of front F_{k+3} can be dominated by a point $P'' \in F_{k+2}$, *i.e.*, $P'' \in F_{k+2}, \forall P''' \in F_{k+3} : P'' \prec P'''$. As dominance relationship is transitive, so all the points of front F_{k+3} can be dominated by point P, *i.e.*,

$$\begin{aligned}(P'' \in F_{k+2}, \forall P''' \in F_{k+3} : P'' \prec P''') \wedge \\ (P' \in F_{k+1}, \forall P'' \in F_{k+2} : P' \prec P'') \wedge \\ (\forall P' \in F_{k+1} : P \prec P') \Rightarrow \\ (\forall P''' \in F_{k+3} : P \prec P''')\end{aligned} \tag{10}$$

In a similar manner, the points of all the upcoming fronts $F_{k+4}, F_{k+5}, \ldots, F_K$ are dominated by $P \in F_k$. Thus, the maximum number of points dominated by a point $P \in F_k$ is obtained by Equation (11).

$$\begin{aligned}\texttt{Dominance_Set}_{\texttt{max}} &= n_{k+1} + n_{k+2} + \cdots + n_K \\ &= \sum_{i=k+1}^{K} n_i\end{aligned} \tag{11}$$

The number of points dominated by a point $P \in F_k$ is basically the cardinality of the dominance set P. Thus, the maximum cardinality of the dominance set of a point $P \in F_k$ is given by Equation (11).

4.1.2 Minimum Cardinality of Dominance Set

As per the definition of non-dominated sorting [fifth condition of the Definition 3], a point $P \in F_k$ is dominated by **at least** one point in its preceding front. It means P can be dominated by **exactly one** point of its preceding front F_{k-1} as well.

Point $P' \in F_{k+1}$ can be dominated by a point $P \in F_k$. Similarly, a point $P'' \in F_{k+2}$ can be dominated by a point $P' \in F_{k+1}$. As the dominance relation is transitive, so point $P'' \in F_{k+2}$ can be dominated by point P, *i.e.*, $(P' \prec P'') \wedge (P \prec P') \Rightarrow (P \prec P'')$.

Point $P''' \in F_{k+3}$ can be dominated by a point $P'' \in F_{k+2}$. As the dominance relation is transitive, so point $P''' \in F_{k+3}$ can be dominated by point P, *i.e.*, $(P'' \prec P''') \wedge (P' \prec P'') \wedge (P \prec P') \Rightarrow (P \prec P'')$.

In a similar manner, one point of all the upcoming fronts $F_{k+4}, F_{k+5}, \ldots, F_K$ can be dominated by point P. Thus, the minimum number of points dominated by a point $P \in F_k$ is obtained by Equation (12).

$$\begin{aligned}\texttt{Dominance_Set}_{\texttt{min}} &= \underbrace{1 + 1 + \cdots + 1}_{\text{Point of fronts } F_{k+1}, F_{k+2}, \ldots, F_K} \\ &= K - (k+1) + 1 \\ &= K - k\end{aligned} \tag{12}$$

The number of points dominated by a point $P \in F_k$ is basically the cardinality of the dominance set P. Thus, the minimum cardinality of the dominance set of a point $P \in F_k$ is given by Equation (12).

Theorem 3. *It is not always* TRUE *that the domination count of $P \in F_k$ is less than the domination count of $P' \in F_{k+1}$. That means $\texttt{dom_count}_P < \texttt{dom_count}_{P'}$ is not always* TRUE.

Proof. The maximum value of domination count of point $P \in F_k$ is $\sum_{i=1}^{k-1} n_i$ and the minimum value is $k-1$ (from Section 3.1). The maximum value of domination count of point $P' \in F_{k+1}$ is $\sum_{i=1}^{k} n_i$ and the minimum value is k.

$$\texttt{dom_count}_P < \texttt{dom_count}_{P'} \tag{13}$$

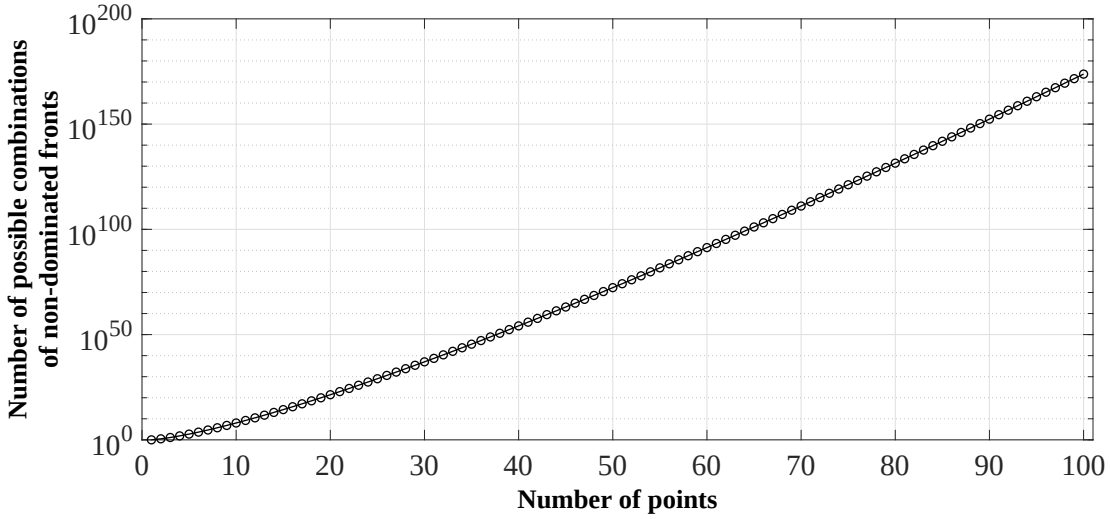


Figure 2: Number of possible combinations of non-dominated fronts for $N(1 \leq N \leq 50)$ points.

For Equation (13) to be TRUE, the following condition should satisfy

$$\max\left(\sum_{i=1}^{k-1} n_i, k-1\right) < \min\left(\sum_{i=1}^{k} n_i, k\right)$$
$$\sum_{i=1}^{k-1} n_i < k \tag{14}$$

As the minimum cardinality of a front can be 1. So Equation (14) will not always satisfy. Thus, $\texttt{dom_count}_P < \texttt{dom_count}_{P'}$ is not always TRUE. □

5 Combinations of Non-dominated Fronts

Here, we obtain all the possible combinations of non-dominated fronts, which can be obtained from N points. Given N points, the number of non-dominated fronts can vary between 1 to N. So, in particular, we first obtain all the possible combinations of non-dominated fronts where the number of fronts is 1. Then, we obtain all the possible combinations of non-dominated fronts where the number of fronts is 2 and so on.

Definition 6 (**Bell Number**). *The total number of possible combinations of non-dominated fronts, which can be obtained from N points is same as N^{th} Bell Number. We denote it as B_N.*

Example 3. *The total number of possible combinations of non-dominated fronts obtained from one point, i.e., B_1 is 1. The value of B_2 is 3. The value of B_3 is 13 and so on. Figure 2 shows the total number of possible combinations of non-dominated fronts B_N for $N(1 \leq N \leq 50)$ points. From this figure, it is clear that these numbers grow exponentially with an increase in N.*

The Bell number B_N gives the total number of possible combinations of non-dominated fronts obtained from N points. However, it does not give any information about the number of possible combinations of non-dominated fronts where the number of fronts is $k(1 \leq k \leq N)$. The *Bell Polynomial* provides this information.

Definition 7 (**Bell Polynomial**). *Bell polynomial $B_N(x_1, x_2, \ldots, x_N)$ is a polynomial that generalizes the Bell number B_N. This polynomial encodes the information related to the ways a set of points are arranged into different non-dominated fronts.*

Example 4. *The first few Bell polynomials are as follows.*

$$\begin{aligned}
B_1(x_1) &= x^1 \\
B_2(x_1, x_2) &= x^1 + 2x^2 \\
B_3(x_1, x_2, x_3) &= x^1 + 6x^2 + 6x^3 \\
B_4(x_1, x_2, x_3, x_4) &= x^1 + 14x^2 + 36x^3 + 24x^4 \\
B_5(x_1, x_2, x_3, x_4, x_5) &= x^1 + 30x^2 + 150x^3 + 240x^4 + 120x^5 \\
B_6(x_1, x_2, x_3, x_4, x_5, x_6) &= x^1 + 62x^2 + 540x^3 + 1560x^4 + 1800x^5 + 720x^6
\end{aligned}$$

In general, Bell polynomial for N points is represented by Eq. (15).

$$B_N(x_1, x_2, \ldots, x_N) = a_N^1 x^1 + a_N^2 x^2 + \ldots + a_N^N x^N \tag{15}$$

The coefficient of kth term in this polynomial, *i.e.*, a_N^k represents the number of possible combinations of non-dominated fronts where the number of fronts is k. The number of monomials in $B_N(x_1, x_2, \ldots, x_N)$ is equal to N. The superscript of x represents the number of non-dominated fronts. The sum of the coefficients of all the terms of $B_N(x_1, x_2, \ldots, x_N)$ provides the Bell number which is given by Eq. (16).

$$B_N = \sum_{k=1}^{N} a_N^k \tag{16}$$

5.1 Process to Obtain Bell Polynomial $B_N(x_1, x_2, \ldots, x_N)$

This process is recursive in nature, *i.e.*, $B_N(x_1, x_2, \ldots, x_N)$ is obtained from $B_{N-1}(x_1, x_2, \ldots, x_{N-1})$, $B_{N-1}(x_1, x_2, \ldots, x_{N-1})$ is obtained from $B_{N-2}(x_1, x_2, \ldots, x_{N-2})$, and so on. $B_{N-1}(x_1, x_2, \ldots, x_{N-1})$ is obtained for $N-1$ points $\{P_1, P_2, \ldots, P_{N-1}\}$ and $B_N(x_1, x_2, \ldots, x_N)$ is obtained for N points $\{P_1, P_2, \ldots, P_N\}$. So, in the process of obtaining $B_N(x_1, x_2, \ldots, x_N)$ from $B_{N-1}(x_1, x_2, \ldots, x_{N-1})$, we have a new point P_N which needs to be added/inserted to all the possible places in the combinations of non-dominated fronts obtained for $N-1$ points. The Bell polynomial for $N-1$ points is given by Eq. (17).

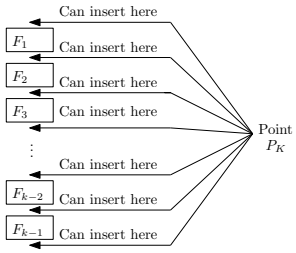
$$B_{N-1}(x_1, x_2, \ldots, x_{N-1}) = a_{N-1}^1 x^1 + a_{N-1}^2 x^2 + \ldots + a_{N-1}^{k-1} x^{k-1} + a_{N-1}^k x^k + \ldots + a_{N-1}^{N-1} x^{N-1} \tag{17}$$

Our aim is to obtain $B_N(x_1, x_2, \ldots, x_N)$ with the help of $B_{N-1}(x_1, x_2, \ldots, x_{N-1})$. As the number of points is N, so the number of fronts varies between 1 and N. First, we find all the possible combinations of non-dominated fronts for N points, where the number of fronts is one. Then, we find all the possible combinations of non-dominated fronts for N points where the number of fronts is two, and so on.

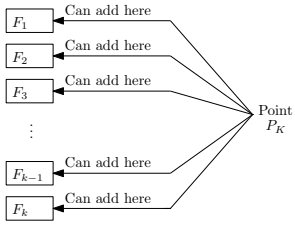
Let us consider that we are finding all the possible combinations of non-dominated fronts for N points where the number of fronts is $k(1 \leq k \leq N)$. In the process of obtaining $B_N(x_1, x_2, \ldots, x_N)$ from $B_{N-1}(x_1, x_2, \ldots, x_{N-1})$, a new point P_N is added to all the possible places in the non-dominated fronts obtained from $B_{N-1}(x_1, x_2, \ldots, x_{N-1})$. We are interested in finding the possible combinations of non-dominated fronts where there are k fronts. We can get these combinations of non-dominated fronts from $B_{N-1}(x_1, x_2, \ldots, x_{N-1})$ where the number of fronts is either $k-1$ or k. For this purpose, $a_{N-1}^{k-1} x^{k-1}$ and $a_{N-1}^k x^k$ from $B_{N-1}(x_1, x_2, \ldots, x_{N-1})$ will be used, as here the number of fronts is $k-1$ and k respectively.

5.2 Process the Combinations of Non-dominated Fronts for $N-1$ Points Where There are $k-1$ Fronts

As there are a_{N-1}^{k-1} possible combinations of non-dominated fronts for $N-1$ points where there are $k-1$ fronts, so we can create a new front with a single point P_N and insert it in between these $k-1$ fronts and also before the first front and after the last front. In this way, we get the combinations of non-dominated fronts where the number of fronts is k. As there are a_{N-1}^{k-1} possible combinations of non-dominated fronts, and the new point can be inserted into any of the $k(= k-2+2)$ places, thus, the number of possible combinations of non-dominated fronts where a new point P_N is considered as a new front is $k \times a_{N-1}^{k-1}$. This scenario is shown in Figure 3(a). In this figure, there are $k-1$ fronts $\{F_1, F_2, \ldots, F_{k-1}\}$ obtained for $N-1$. We want to have k fronts which can be obtained for N points. So, point P_N is considered as a new front and can be inserted in between any of the $k-1$ fronts and also before the first front F_1 and after the last front F_{k-1}.



(a) A point creates a new front and can be inserted in between any of the existing $k-1$ fronts and also before the first front and after the last front.



(b) A point can be inserted into any of the existing k fronts.

Figure 3: The process to find all the possible combinations of non-dominated fronts for N points with the help of the non-dominated fronts which have been obtained for $N-1$ points. Only those combinations of non-dominated fronts are obtained where there are k fronts.

5.3 Process the Combinations of Non-dominated Fronts for $N-1$ Points Where There are k Fronts

As there are a_{N-1}^k possible combinations of non-dominated fronts for $N-1$ points when there are k fronts, so we can add a point P_N to any of the existing k fronts. In this way, we get the combinations of non-dominated fronts where the number of fronts is k. As there are a_{N-1}^k possible combinations of non-dominated fronts, and the point P_N can be added to any of these k fronts, thus, the number of possible combinations of non-dominated fronts where a new point is added to one of the existing fronts is $k \times a_{N-1}^k$. This scenario is shown in Figure 3(b). In this figure, there are k fronts $\{F_1, F_2, \ldots, F_k\}$ obtained for $N-1$ points. We want to have k fronts for N points. So, point P_N can be added to any of the k fronts.

Hence, the number of possible combinations of non-dominated fronts for N points when there are k number of fronts is $k \times a_{N-1}^{k-1} + k \times a_{N-1}^k = k \times \left(a_{N-1}^{k-1} + a_{N-1}^k\right)$.

Till now we have discussed the method to find all the possible combinations of non-dominated fronts for N points with k number of fronts using the combinations of non-dominated fronts for $N-1$ points. However, our main aim is to find all the possible combinations of non-dominated fronts for N points with 1 to N fronts. So, the value of k should vary between 1 to N. However, when the value of k is 1, then $k-1$ will be negative so we ignore $k-1$ in this case. Also, when the value of k is N, then we do not have the combinations of non-dominated fronts for $N-1$ points where there are N fronts (from $N-1$ points, the maximum number of fronts which can be obtained is $N-1$), so we ignore k in this case. Next, we describe how to handle these two boundary cases.

5.4 Handling the Situation When There is a Single Front

The non-dominated front obtained for $N-1$ points when there is a single front, is $\{P_1, P_2, \ldots, P_{N-1}\}$. Thus, when we need to find the possible combinations of non-dominated fronts for N points with a single front, then P_N is added to $\{P_1, P_2, \ldots, P_{N-1}\}$.

5.5 Handling the Situation When There are N Fronts

From $N-1$ points we cannot get N fronts so we only consider the combinations of non-dominated fronts for $N-1$ points where there are $N-1$ fronts and obtain all the possible combinations of non-dominated fronts.

We have seen the approach to obtain $B_N(x_1, x_2, \ldots, x_N)$ from $B_{N-1}(x_1, x_2, \ldots, x_{N-1})$. Thus, we can get B_N from $B_N(x_1, x_2, \ldots, x_N)$ by adding the coefficients of all the terms in the polynomial as in Eq. (16).

The non-dominated front polynomial $B_N(x_1, x_2, \ldots, x_N)$ can be obtained using $B_{N-1}(x_1, x_2, \ldots, x_{N-1})$ (given in Eq. (17)) with the help of Eq. (18).

$$B_N(x_1, x_2, \ldots, x_N) = a_{N-1}^1 x^1 + \left[\sum_{k=2}^{N-1} k(a_{N-1}^{k-1} + a_{N-1}^k)x^k\right] + N \cdot a_{N-1}^{N-1} x^N \tag{18}$$

Given $B_{N-1}(x_1, x_2, \ldots, x_{N-1})$, the coefficient of the $k^{th}(1 < k < N)$ term in $B_N(x_1, x_2, \ldots, x_N)$ is obtained by Eq. (19). The process to obtain the coefficients of all the N terms in $B_N(x_1, x_2, \ldots, x_N)$ from the coefficients of

$B_{N-1}(x_1, x_2, \ldots, x_{N-1})$ is shown in Figure 4.

$$a_N^k = k\left[a_{N-1}^{k-1} + a_{N-1}^k\right] \qquad \text{for } 2 \leq k \leq N-1 \tag{19}$$

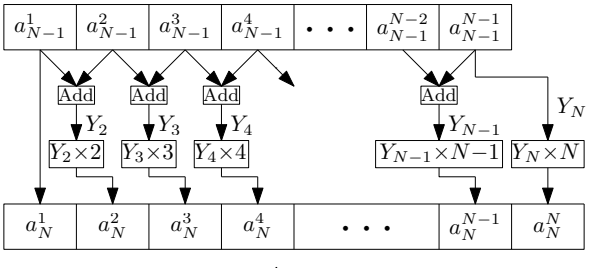


Figure 4: The process to obtain the coefficients of all the terms in $B_N(x_1, x_2, \ldots, x_N)$ from the coefficients of $B_{N-1}(x_1, x_2, \ldots, x_{N-1})$.

Using the technique to obtain the coefficients of all the terms in $B_N(x_1, x_2, \ldots, x_N)$ from $B_{N-1}(x_1, x_2, \ldots, x_{N-1})$, we can produce a triangle as shown in Figure 5. Let us call it Ndf (non-dominated front) triangle. The sum of values of all the columns of the Nth row tells us the Bell number B_N. Figure 5 shows the ndf triangle for $N(1 \leq N \leq 10)$ points.

											Bell Number
$N = 1$	1										1
$N = 2$	1	2									3
$N = 3$	1	6	6								13
$N = 4$	1	14	36	24							75
$N = 5$	1	30	150	240	120						541
$N = 6$	1	62	540	1560	1800	720					4683
$N = 7$	1	126	1806	8400	16800	15120	5040				47293
$N = 8$	1	254	5796	40824	126000	191520	141120	40320			545835
$N = 9$	1	510	18150	186480	834120	1905120	2328480	1451520	362880		7087261
$N = 10$	1	1022	55980	818520	5103000	16435440	29635200	30240000	16329600	3628800	102247563

Figure 5: Ndf triangle which shows the Bell number that can be obtained from $N(1 \leq N \leq 10)$ points.

6 Questions

- Time to obtain the dominance relationship.
- Maximum/Minimum number of objective value comparisons to obtain the dominance relationship and when will it happen.
- Maximum/Minimum value of domination count for a point if the front is known/unknown and when will it happen.
- Benefit of sorting the points based on a particular objective in case of efficient non-dominated sort.
- Maximum/Minimum number of comparisons between the points (also known as dominance comparisons) in Naive approach.
- Maximum/Minimum number of dominance comparisons in Fast Non-dominated Sort.
- Maximum number of dominance comparisons in Efficient Non-dominated Sort – Sequential and Binary version both.
- When the number of dominance comparisons will be maximum in case of Efficient Non-dominated Sort – Sequential version.
- When the number of dominance comparisons will be maximum in case of Efficient Non-dominated Sort – Binary version.
- How Fast Non-dominated Sort is related to the Naive approach.
- Worst and best case time complexity of pre-sorting and when will these two scenarios occur.

- How tie is resolved in pre-sorting.
- In pre-sorting, what will happen if we do not break the tie and only consider the first objective? Can it will still give the same benefit that it was giving when we break the tie considering other objectives.
- Maximum/Minimum number of fronts which can be obtained by any non-dominated sorting algorithm.
- Given 5 points, these points can be sorted in how many different ways. Why is this less than the possible clustering arrangements obtained from 5 points?
- As non-dominated sorting is also a sorting algorithm, why cannot we directly use merge-sort, Quick-sort, etc.?
- Given three points P, Q and R. Give a scenario, where P and Q are non-dominated, Q and R are non-dominated but P and R are not non-dominated.
- If the dominance relationship (a point dominates another) between the points is represented using a Graph, then obtain the maximum/minimum number of edges in the graph. When will it happen.

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